COURSE SECTION INFORMATION (CSI)

CST8507: Natural Language Processing

Artificial Intelligence Software Development

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Course Section:01

Academic Year:2026

Term: Winter 2026

Academic Level: 2

Learning Resources

Recommended Textbook:

- Jurafsky, D. & Martin, J. H. (2024) *Speech and Language Processing (3rd ed. draft)*. https://web.stanford.edu/~jurafsky/slp3/ed3book.pdf
- Bird, S., Klein, E., & Loper, E. *Natural Language Processing with Python.* (O'Reilly 2009, website 2018), http://www.nltk.org/book/

Hardware and Software:

Python: latest version
Anaconda: latest version
Google Colab

Evaluation Breakdown

Assessment	Value	CLRs
Lab (4x 5%)	20%	1, 2,3, 4,5
Assignment (Due Week 6)	10%	1, 2,3, 4
Group Project(Due Week 12)	20%	1, 2,3, 4,5,6
Total Practical	**50%**	
Midterm (1 x 15%, Week 7)	15%	1, 2,3, 4
Final Exam (1 x 25%, Week 15)	25%	1, 2,3, 4,5,6
Class Participation (5 out of 8)	10%	1, 2,3, 4,5
Total Theory	**50%**	

Learning Schedule (subject to change with notification)

Date	Weekly Theme and Learning Outcomes	Learning Activities	Assessments (%)	Resources	CLRs
Week 1 **12/1/2026**	Course outline and logistics Introduction and Overview • Motivations • Natural Language Processing Applications • Natural Language Processing Python Libraries	• Lecture • **Hybrid Activity (1)** • **Lab1**		• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	1, 2
Week 2 **19/1/2026**	Text preprocessing and exploratory analysis • Regular expression • Tokenization • Stemming • Removing stop words • Part Of Speech tagging (POS)	• Lecture • **Hybrid Activity (2)**	Participation 1 (1.25%).	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	1,2
Week 3 **26/1/2026**	Text Vectorization I • Word meaning (basic terminology) • Word similarity • Frequency based Text representation (TF-IDF)	• Lecture • **Hybrid Activity (3)** • Lab 2 • Assignment 1	Participation 2(1.25%). **Lab 1 Demo (5%)**	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	2,3

Week 4 **2/2/2026**	Word Embedding • CBOW • Skip-Gram • GloVe • Doc2vec • FastText	• Lecture • Hybrid Activity (4)	Participation 3 (1.25%).	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	3,4
Week 5 **9/2/2026**	Deep Learning Architectures for Sequence Processing • RNN as language Model • LSTM and tree neural network • Applications of Deep Learning in NLP • Sequence to Sequence Models	• Lecture • **Hybrid Activity (5)** • Lab3	Participation 4 (1.25%). **Lab 2 Demo (5%)**	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	1
Week 6 **16/2/2026**	**sequence to sequence Models** • The encoder-decoder framework • Self-Attention mechanisms • Introduction to Transformer-based Language Models	• Lecture • **Hybrid Activity (6)** • **Project** Proposal & Group Formation	Participation 5(1.25%). **Assignment 1 demo (10%)**	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	3,4
Week 7 **23/2/2026**	**Midterm**	**Midterm**	**Midterm (15%)**		1,2,3,4
Week 8 **2/3/2026**	BREAK	BREAK	BREAK	BREAK	

Week 9 **9/3/2026**	Transformer Architecture • Positional encoding • Q,K, and V • Multi-head attention • Masking		Participation 6(1.25%). **Lab 3 Demo (5%)**	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	4,5
Week 10 **16/3/2026**	Bert & Classification for text analysis • BERT Architecture • BERT Tokenization • BERT variants • BERT for text classification Introduction to Question Answering systems • Extractive Question Answering (Reading Comprehension) • Open Domain Vs closed domain Question Answering • Retriever-Reader Architecture	• Lecture • **Hybrid Activity (8)** • **Project**	Participation 7 (1.25%). **Project proposal demo (3%)**	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	1,2,3,4
Week 11 **23/3/2026**	Introduction to LLM • LLMs Components • How ChatGPT trained • Limitations of LLMs • Retrieval Augmented Generation **RAG** works • LangChian	• Lecture • **Hybrid Activity (9)** • **Project**	• Participation 8(1.25%). **Lab 4 Demo (5%)**	• Lecture Slides • Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	4,5

Week 12 **30/3/2026**	Model Compression • Model Compression techniques Introduction to Prompt Engineering • Types of Prompt Engineering • Prompt Engineering best practice	Lab: Working with Project • **Hybrid Activity (10)**		• Other Topic specific resources and book chapters are included in the course materials available on Brightspace.	1,2,4,6
Week 13 **6/4/2026**	Sentiment Analysis Clustering for Text Similarity • Document clustering models • Latent Dirichlet Allocation • Non-Negative Matrix Factorizationon	• Project Student Presentations	**Project Demo (20%)**		1,2,3 4,5,6
Week 14 **13/4/2026**	○ Recovery Week	• Project ○ Student Presentations	**Project Demo (20%)**		1,2,3 4,5,6
Week 15	**Final Exam (25%)**				

Other Important Information

- The lectures will cover the course material. Students are expected to attend all the lectures.
- **You must get at least 50% in practical (Lab activities, Assignment, and project) and at least 50% in theory (Participation, midterm, and final exam) to pass the course. Simply getting 50% overall without meeting the 50% min in each theory and practical is not sufficient to pass the course.**
- Labs**, and Assignments** must be demonstrated immediately **after the due date** and according to their scheduled weeks (as stated in the CSI). **Missing Demo on the specified date may result in zero marks.**
- Lab and assignment demos will take place during your scheduled lab session. During the demo, your lab instructor may ask you to complete a small additional task or answer one or two questions to confirm that the work is your own. If you're unable to do so, the assignment may receive a zero mark.
- **The demo is not for evaluation;** your lab professor will assess your submitted work on Brightspace to ensure that it meets the requirements of the labs or assignments.
- Demonstrating something that does not match what you submitted on Brightspace would result in a zero mark.
- The official submission deadline for all labs and assignments is **Friday night (11:59 PM).** To accommodate all learners, I allow submissions until **Sunday night (11:59 PM) without any penalties**. Please take advantage of this grace period to manage your workload effectively.

I strongly encourage everyone to aim for the official deadline on Friday night to avoid last-minute stress.

- ❖ All labs and assignment 1 are individual work.
- ❖ Project is group work. Please stick with the following guidelines
  - **Group Size**: Each group for Project should consist of a maximum of **Two** members.
  - Each student is responsible for enrolling themselves in a designated group by the specified deadline. Failure to enroll in a group by the due date will result in a course penalty being applied.
- ❖ Take advantage of your lab time to ask your questions, don't wait for the last day before the due date. **Technical questions should be addressed during lab sessions not by email.**
- ❖ For every lab and assignment, specific submission instructions are provided. It is crucial that you **carefully follow these instructions**. Failing to adhere to the submission instructions, may result in receiving a grade of zero for that lab or assignment.
- ❖ Late submissions for Labs and Assignments will incur a penalty of 10% per day beyond the due date. The penalty structure is as follows:
  - 1st day: -20%
  - 2nd day: -40%
  - 3rd day: -100%, meaning zero.
- ❖ If you encounter difficulties such as submission issues, or Brightspace related problems you must notify your professor by email ***before*** the assignment deadline and include your completed work as an attachment to that email. Requests made after the due date cannot be accommodated.
- ❖ You are responsible for uploading the correct deliverable to the correct upload link.
- ❖ After submitting your work, you can review your submissions on Brightspace. It's essential that you verify you haven't uploaded the wrong file or an empty document. If your lab professor does not receive the correct file, your assessment will be marked as zero, and you will not have the opportunity to resubmit.

- ❖ Citations and References are needed in all documents you create and pass in when you use resources external from the course handouts. This includes placing citations and references within source code using IEEE style as programmer comments.
- ❖ Copy and pasting huge portions of documents (including source code) from other authors or generative AI technology will not earn you marks even if properly cited and referenced.
- ❖ **No late submission for Project.**
- ❖ Starting from the week 2, students will be required to take a Participation that assesses the topics covered in the preceding lecture + Hybrid. There will be 8 Participations in total. They should take approximately ten minutes to complete. These Participations serve as a reinforcement of the week's content and are integral to the learning process. Participation will be proctored during the lecture. Five best Participations will be counted into final score. Each Participation will worth 2% of final score.
- ❖ Any participation submitted **without attending the corresponding lecture** will receive a zero mark.
- ❖ Midterm and final Exams will be in person, on paper and closed book.

- Expect email reply is within two business days. Therefore, you need to plan your emails accordingly.
    - **Please, Include in your email:**
        - Your name
        - Course name and code
        - Section No.
- Use your Algonquin College student email account. I may ignore emails sent from other sources (Gmail, Hotmail etc.) to protect student privacy and security.

- The educational value of this course lies not only in the mastery of course content but also in the process of engaging first with the material. The work you do in this course is intended to strengthen original thought, critical thinking, and individual problem-solving skills. The use of generative AI to complete work compromises the learning process and the achievement of course learning requirements.

  Under Algonquin College Policy AA48 – Academic Integrity, "Academic work submitted by learners is evaluated on the assumption that the work presented by the learner is their own" and defines contract cheating as "[a] third-party completing work, with or without payment, for a learner, who then submits the work as their own, where such input is not permitted." Using content generated by AI and claiming it as your own work is considered contract cheating. Violations of these expectations will be brought forward as instances of academic misconduct under this policy.